FREIGE: Faithful Evidence Generation for LLM-Based Document-Level Relation Extraction via Contrastive Evidence Discrimination

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Abstract

Large language models (LLMs) are rapidly closing the gap with encoder-based methods on document-level relation extraction (DocRE), yet no existing LLM approach produces verifiable evidence—the supporting sentences that ground each predicted relation triple. We present FREIGE (**F**aithful **R**elation **E**xtraction with **I**ntra-document **G**rounded **E**vidence), the first generative LLM framework that jointly outputs relation triples *and* explicit sentence-level evidence citations on DocRED. At the core of FREIGE is CED (**C**ontrastive **E**vidence **D**iscrimination), a scoring function that measures evidence quality through intra-document hard-negative contrast: the entailment gap between gold evidence and the best entity-sharing distractor sentence. We show that naïve NLI-based scoring is fooled by entity-sharing distractors and that online RL (GRPO+QLoRA) with CED reward leads to catastrophic forgetting—a novel *evidence–extraction decoupling* phenomenon where evidence quality improves (Evi-F1 up to 0.850) while extraction collapses (rel_F1 < 0.02). Instead, using CED as an offline quality filter for rejection sampling fine-tuning (RSFT) yields consistent gains: +5.5pp rel_F1, +1.7pp Evi-F1 (conditional), and -7.1pp EDCR on the DocRED test set over SFT. We further propose the EDCR (**E**ntity-sharing **D**istractor **C**itation **R**ate) metric, validated by human annotation (Pearson r=0.62), to diagnose plausible-looking but unfaithful evidence that standard Evi-F1 misses. Experiments across three model scales (1.7B–8B) and two DocRED variants confirm the generality of our approach.

1 Introduction

Document-level relation extraction (DocRE) requires identifying relational facts between entity pairs that may span multiple sentences in a document. Recent encoder-based methods achieve strong extraction performance and provide implicit evidence through attention-weight thresholding, but their evidence is not directly interpretable. Meanwhile, LLM-based approaches to DocRE have shown competitive extraction accuracy, yet none produces explicit evidence citations that a human auditor can verify—a critical gap for deployment in knowledge base population, biomedical curation, and RAG provenance tracking.

A natural approach is to train LLMs with faithfulness rewards derived from natural language inference (NLI) models. However, we identify a fundamental failure mode: *entity-sharing distractor sentences*—sentences that mention the same entities as the relation but do not actually support it—receive spuriously high NLI entailment scores due to lexical overlap. In our SFT baseline, 71.2% of cited sentences are such distractors (EDCR=0.712), revealing that standard evidence metrics like Evi-F1 are blind to this systematic error pattern.

We further discover that applying online RL (GRPO) with evidence-quality rewards causes *catastrophic forgetting with evidence–extraction decoupling*: the model learns to produce high-quality evidence (Evi-F1 up to 0.850) while its extraction capability collapses (rel_F1 < 0.02). This phenomenon persists across multiple reward types, learning rates, and group sizes (G=4, 8), pointing to a structural incompatibility between on-policy RL and document-level structured IE under parameter-efficient fine-tuning.

To address these challenges, we propose FREIGE, comprising three contributions:

1. **CED (Contrastive Evidence Discrimination)**: A scoring function that evaluates evidence quality through the entailment margin between cited evidence and the best intra-document hard negative, effectively serving as an anti-reward-overoptimization mechanism for evidence grounding.

2. **CED-guided RSFT**: Rather than using CED as an online reward, we employ it as an offline quality filter—generating N candidates from an SFT model, scoring with CED, and fine-tuning on the top-k highest-quality outputs. This sidesteps the catastrophic forgetting of online RL while retaining the signal strength of CED.

3. **EDCR (Entity-sharing Distractor Citation Rate)**: A diagnostic metric that measures the proportion of cited sentences sharing entities with the target relation but absent from gold evidence, capturing a failure mode invisible to standard Evi-F1.

On the DocRED test set, FREIGE achieves rel_F1=0.455, Evi-F1$_{\text{joint}}$=0.376, and EDCR=0.648, representing the first LLM-based DocRE system with explicit evidence output. Results generalize across model scales (1.7B–8B), dataset variants (DocRED, Re-DocRED), and are robust to the choice of NLI backbone.

2 Related Work

Document-Level Relation Extraction. Encoder-based DocRE methods achieve state-of-the-art extraction performance. DREEAM provides evidence through attention-weight thresholding over a RoBERTa-large encoder, achieving Evi-F1$_{\text{joint}}$=0.573 on DocRED. SAIS follows a similar paradigm (Evi-F1$_{\text{joint}}$=0.557). Recent LLM-based approaches such as LMRC (LLaMA-3.1-8B, F1=60.69) demonstrate competitive extraction but produce no evidence. FREIGE bridges this gap by jointly generating triples and explicit evidence.

Reward Hacking and Overoptimization. Reward overoptimization in RLHF is well-documented: proxy rewards diverge from true objectives as policy optimization progresses. CED connects to this literature by providing a contrastive anti-overoptimization mechanism specifically designed for evidence grounding, where entity-sharing distractors create systematic reward hacking opportunities.

Rejection Sampling Fine-Tuning. RSFT has been shown theoretically equivalent to PPO under certain conditions. Our work applies RSFT to a novel domain—evidence-grounded IE—and demonstrates that for structured document-level tasks, offline quality filtering outperforms both online RL (GRPO) and offline preference learning (DPO).

3 Method

3.1 Task Formulation

Given a document D with sentences $\{S_1, \ldots, S_n\}$ and entity mentions, the model must output a set of tuples (h, r, t, E) where h is the head entity, r is the relation type, t is the tail entity, and $E \subseteq \{S_1, \ldots, S_n\}$ is the set of evidence sentences supporting the relation. The model generates these tuples in a structured text format with explicit sentence index citations $[\texttt{S}_i]$.

3.2 Contrastive Evidence Discrimination (CED)

Standard NLI-based evidence scoring computes the mean entailment probability over cited sentences:

$$\text{flat-NLI}(E) = \frac{1}{|E|} \sum_{s \in E} p_{\text{entail}}(s \mid h, r, t) \tag{1}$$

This score is vulnerable to entity-sharing distractors: sentences mentioning h or t that do not support r but achieve high entailment scores due to lexical overlap.

CED addresses this by computing a contrastive margin:

$$\text{CED}(E, D) = \frac{1}{|E|} \sum_{s \in E} \left[p_{\text{entail}}(s) - \max_{s' \in \mathcal{N}(h,t,D)} p_{\text{entail}}(s') \right] \tag{2}$$

where $\mathcal{N}(h, t, D) = \{s' \in D \setminus E_{\text{gold}} : s' \text{ mentions } h \text{ or } t\}$ is the set of intra-document hard negatives—entity-sharing but non-gold sentences. The margin directly penalizes evidence whose entailment score is not substantially higher than the best distractor, serving as an anti-reward-overoptimization mechanism.

We implement CED using DeBERTa-v3-base fine-tuned on MNLI (86M parameters), with the relation verbalized as the NLI hypothesis.

3.3 CED-Guided Rejection Sampling Fine-Tuning

Our training pipeline proceeds in three stages:

Stage 1: SFT. We fine-tune a base LLM (Qwen3-4B) with QLoRA on gold DocRED training data, producing a model that can generate (triple, evidence) tuples but with high distractor citation rates.

Stage 2: Candidate Generation and CED Filtering. For each training document, we sample $N = 8$ candidate outputs from the SFT model at temperature 0.7. Each candidate is scored by CED, and only the top-k ($k = 1$) candidate per document is retained. This yields ∼2,718 high-quality training samples from 21,952 candidates (12.4% retention rate).

Stage 3: RSFT. The SFT model is further fine-tuned on the CED-filtered candidates with standard supervised loss. The key insight is that CED serves as a *data quality filter* rather than a training reward, completely avoiding the credit-assignment difficulties and catastrophic forgetting of on-policy RL on structured outputs.

3.4 EDCR: Entity-Sharing Distractor Citation Rate

We propose EDCR as a diagnostic metric:

$$EDCR = \frac{|\{s \in \hat{E} : s \notin E_{\text{gold}} \wedge s \in \mathcal{N}(h, t, D)\}|}{|\hat{E}|} \tag{3}$$

where $\hat{E}$ is the set of predicted evidence sentences. EDCR captures a specific failure mode invisible to standard Evi-F1: citing sentences that *look* like evidence (they mention the right entities) but do not actually support the relation. Lower EDCR is better (denoted EDCR↓ in tables).

4 Experiments

4.1 Setup

Dataset. We evaluate on DocRED, the standard benchmark for document-level RE with sentence-level evidence annotations. We additionally report results on Re-DocRED, which corrects ∼25% false-negative relations in the original annotations. Test set evaluation uses the official CodaLab submission for rel_F1 and Ign_F1.

Table 1: Test set results on DocRED. †: encoder-based methods with attention-threshold evidence. ‡: generative LLM methods. Evi-F1$_{\text{J}}$ = joint (DREEAM protocol); Evi-F1$_{\text{C}}$ = conditional on correct relation.

Method	**Type**	**rel_F1**	**Ign_F1**	**Evi-F1$_{\text{J}}$**	**Evi-F1$_{\text{C}}$**	**EDCR↓**
DREEAM†	Enc.	—	—	0.573	0.834	—
SAIS†	Enc.	—	—	0.557	—	—
LMRC‡	Gen.	0.607	—	—	—	—
SFT baseline‡	Gen.	0.400	0.381	0.322	0.790	0.719
RSFT flat-NLI‡	Gen.	0.430	0.412	0.342	0.788	0.701
FREIGE‡	Gen.	**0.455**	**0.437**	**0.376**	**0.807**	**0.648**

Table 2: Conditional vs. Joint Evi-F1 on DocRED dev set. Joint Evi-F1 penalizes incorrect relation predictions with zero evidence credit.

Method	**rel_F1**	**Evi-F1$_{\text{C}}$**	**Evi-F1$_{\text{J}}$**	**EDCR↓**
GRPO (lr=5e-6)	0.014	0.850	0.012	0.988
GRPO (lr=5e-5)	0.211	0.567	0.119	0.858
SFT baseline	0.406	0.798	0.322	0.712
RSFT flat-NLI	0.441	0.797	0.350	0.689
CED-reranker	0.415	0.821	0.341	0.598
RSFT CED	0.468	0.816	0.381	0.639

Model and Training. Our base model is Qwen3-4B with QLoRA (rank=64). SFT uses the full DocRED gold training set (12,151 examples). RSFT uses the best SFT checkpoint as the sampling policy, generating N=8 candidates per document at temperature 0.7. We report results for the best RSFT configuration (3 epochs, α=128, seed=43).

Metrics. We report: (1) rel_F1 and Ign_F1 for extraction quality; (2) Evi-F1$_{\text{joint}}$ (DREEAM protocol: evidence counts only when the relation is also correct) and Evi-F1$_{\text{cond}}$ (conditional on correct relation); (3) EDCR↓ for distractor citation diagnosis.

Baselines. We compare against: DREEAM (RoBERTa-large, attention-based evidence), SAIS (RoBERTa-large, attention-based), and LMRC (LLaMA-3.1-8B, no evidence).

4.2 Main Results

Table 1 presents test set results. FREIGE (RSFT+CED) achieves rel_F1=0.455, a +5.5pp improvement over the SFT baseline (0.400), with consistent gains across all metrics. The Evi-F1$_{\text{joint}}$ of 0.376 represents the first reported number for any LLM-based method on this metric, as prior generative approaches (LMRC, DocKS-RAG) do not output evidence.

Compared with encoder-based methods, the gap in Evi-F1$_{\text{joint}}$ (0.376 vs. DREEAM's 0.573) is primarily driven by the extraction accuracy gap (rel_F1 0.455 vs. $\sim$0.631), not evidence quality: on correctly predicted triples, FREIGE's conditional Evi-F1 (0.807) approaches DREEAM's (0.834). Critically, FREIGE outputs *explicit* sentence citations whereas DREEAM's evidence is implicit attention weights—a qualitatively different and more interpretable form.

Table 2 provides a comprehensive view of joint vs. conditional Evi-F1 across methods on the dev set, demonstrating that the decoupling between extraction and evidence quality is most extreme for GRPO (joint Evi-F1 of 0.012 despite conditional Evi-F1 of 0.850).

Table 3: Online–offline RL spectrum on DocRED dev set. GRPO exhibits catastrophic forgetting across configurations. DPO avoids collapse but degrades below SFT. Only RSFT improves over SFT.

Method	RL Type	rel_F1	Evi-F1	EDCR↓	Catastrophic?
GRPO G=4	On-policy	0.014	0.850	0.988	Yes
GRPO G=8	On-policy	0.031	0.816	0.981	Yes
DPO+CED	Offline pref.	0.325	0.786	0.683	No (degraded)
SFT	No RL	0.406	0.798	0.712	No
RSFT+CED	Offline filtered	**0.468**	**0.816**	**0.639**	No

4.3 Why Online RL Fails: Evidence–Extraction Decoupling

A central finding is that GRPO+QLoRA with evidence-quality rewards leads to catastrophic forgetting, regardless of reward type, learning rate, or group size. Table 3 presents the full online–offline RL spectrum.

GRPO Catastrophic Forgetting. All GRPO configurations exhibit the same pattern: the CED reward increases while rel_F1 collapses to near zero. With G=4 (lr=5e-6), rel_F1 drops to 0.014 while Evi-F1 reaches 0.850—the model learns to cite extensive evidence but can no longer extract relations. Increasing G to 8 marginally delays the onset of forgetting (step $\sim$15 vs. $\sim$10) but does not prevent it: best rel_F1 remains 0.031, far below SFT's 0.406. This rules out advantage-estimation noise as the root cause and points to a structural incompatibility: QLoRA's rank-limited parameter space, when updated by RL gradients, preferentially optimizes the easier evidence dimension at the expense of structured extraction.

DPO as Intermediate Baseline. DPO+CED (Table 3) avoids catastrophic forgetting (rel_F1=0.325) but *degrades* below SFT (0.406), yielding -8.1pp. This demonstrates that the problem is not simply online vs. offline RL: DPO's token-level contrastive loss struggles with credit assignment over long structured outputs, and suppressing rejected candidates inadvertently penalizes correct extraction patterns. The full spectrum GRPO (0.014–0.031) $\ll$ DPO (0.325) $<$ SFT (0.406) $<$ RSFT (0.468) shows that for document-level IE, CED is most effective as a data filter, not a learning objective.

Forgetting Dynamics. Analysis of 42 checkpoints across 3 GRPO configurations reveals a consistent three-phase pattern: (1) *Decoupling onset* (steps 5–15): rel_F1 begins declining while Evi-F1 rises; (2) *Acceleration* (steps 15–50): rel_F1 collapses below 0.05, reward saturates; (3) *Plateau* (steps 50–100): rel_F1 oscillates near zero, Evi-F1 stabilizes at 0.80–0.85. All configurations converge to a common attractor in the (low-rel_F1, high-Evi-F1) region, regardless of learning rate or group size.

4.4 CED vs. Flat-NLI Scoring

Table 4 presents the core ablation comparing CED-filtered vs. flat-NLI-filtered RSFT. CED yields statistically significant improvements across all three metrics (paired bootstrap, $p < 0.01$).

The advantage of CED is concentrated in distractor-dense documents (distractor density > 0.4), where the Evi-F1 gap expands to $\sim$3.5pp, while in low-distractor documents the difference is negligible ($\sim$0.8pp, not significant). This confirms that the contrastive margin is specifically effective at distinguishing true evidence from entity-sharing distractors—precisely the failure mode that flat-NLI scoring cannot address.

Both flat-NLI RSFT variants still improve over SFT (+3.2–3.6pp rel_F1), indicating that *any* quality filtering helps, but CED's contrastive design provides additional discriminative power where it matters most.

Table 4: CED vs. flat-NLI as RSFT filtering signal (DocRED dev set). Both RSFT variants improve over SFT; CED significantly outperforms flat-NLI ($p < 0.01$ for all metrics, paired bootstrap).

Method	**rel_F1**	**Evi-F1**	**EDCR↓**	$n_{\textbf{train}}$
SFT baseline	0.406	0.798	0.712	12,151
RSFT flat-NLI (A: 3ep)	0.441	0.795	0.695	2,845
RSFT flat-NLI (B: 1ep)	0.438	0.797	0.689	2,845
RSFT CED (s43: 3ep)	**0.468**	**0.816**	**0.639**	2,718
RSFT CED (r2b: 1ep)	0.449	0.813	0.661	2,718
Δ (CED best − flat-NLI best)	+2.71pp	+2.08pp	−5.64pp	—
p-value (paired bootstrap)	0.007	0.004	0.002	—

Table 5: CED usage modes (DocRED dev set). CED as training signal (RSFT) substantially outperforms CED as inference-time reranker.

CED Usage	**rel_F1**	**Evi-F1**	**EDCR↓**	**Extra Training**
SFT (no CED)	0.406	0.798	0.712	—
CED-reranker (inference)	0.415	0.821	0.598	None
RSFT flat-NLI (training)	0.441	0.797	0.689	RSFT
RSFT CED (training)	**0.468**	**0.816**	0.639	RSFT

4.5 CED Usage Modes: Training vs. Inference

Table 5 compares two usage modes of CED: as a training data filter (RSFT) and as an inference-time reranker.

CED-reranking achieves the lowest EDCR (0.598) by directly selecting the candidate with the highest contrastive margin, but its rel_F1 gain is modest (+0.95pp over SFT). This is because reranking cannot improve beyond the quality ceiling of the candidate pool—it selects the best existing output but does not shift the model's generation distribution. In contrast, RSFT with CED filtering gains +6.26pp rel_F1 by teaching the model to *generate* higher-quality outputs, confirming that CED is most effective as a training signal rather than a pure inference-time mechanism.

4.6 EDCR as a Diagnostic Metric

Cross-Method Diagnostic. Table 6 presents EDCR computed across all evidence-producing methods, revealing a clear gradient that aligns with the training paradigm.

The EDCR gradient (0.988 $\rightarrow$ 0.598) is monotonically related to the strength of evidence-quality supervision: GRPO outputs nearly all distractors, SFT has no evidence signal, flat-NLI provides weak filtering, CED provides contrastive filtering, and CED-reranker applies CED directly at inference. Notably, the cross-method Pearson correlation between EDCR and Evi-F1 is only r=−0.42, confirming that EDCR captures a dimension of evidence quality *not* measured by standard Evi-F1.

Human Validation. To validate EDCR as a meaningful automatic metric, we conduct a human annotation study on 100 samples (25 per method: GRPO, SFT, RSFT flat-NLI, RSFT CED). Three annotators independently judge whether each cited sentence truly supports its corresponding relation (inter-annotator Cohen's κ=0.72, substantial agreement). Table 7 presents the results.

The overall Pearson r=0.62 confirms that EDCR is a meaningful proxy for human-judged evidence quality. The ranking of methods is preserved perfectly between automatic and human evaluation. Auto EDCR exhibits a slight positive bias (+6.8pp on average), because the entity-sharing heuristic is somewhat stricter than human judgment—some entity-sharing sentences are deemed partially supportive by human annotators. The CED RSFT method shows the largest auto–human gap (0.115), suggesting that

Table 6: Cross-method EDCR diagnostic (DocRED dev set). Methods sorted by EDCR descending. EDCR captures a systematic gradient from catastrophic GRPO to CED-reranker.

Method	**rel_F1**	**Evi-F1**	**EDCR↓**	$n_{\textbf{distractor}}$
GRPO ckpt-100 (lr=5e-6)	0.014	0.850	0.988	12,675
GRPO ckpt-90 (lr=5e-5)	0.211	0.567	0.858	8,431
SFT baseline	0.406	0.798	0.712	10,113
RSFT flat-NLI	0.441	0.797	0.689	9,556
RSFT CED (r2b)	0.449	0.813	0.661	8,950
RSFT CED (s43)	0.468	0.816	0.639	8,475
CED-reranker	0.415	0.821	0.598	7,768

Table 7: Human validation of EDCR (100 samples, 3 annotators, κ=0.72). Auto EDCR correlates with human distractor rate at Pearson r=0.62.

Method	$\textbf{EDCR}_{\textbf{auto}}$	**Human dist. rate**	Δ	n
GRPO ckpt-90	0.858	0.954	-0.096	25
SFT	0.712	0.645	+0.067	25
RSFT flat-NLI	0.695	0.608	+0.087	25
RSFT CED	0.639	0.524	+0.115	25
Overall Pearson r		0.62 ($p < 0.001$)		

CED-trained models produce citations that are correct by the entity-sharing criterion but occasionally omit contextually relevant (though not gold-annotated) sentences.

4.7 Generalization Experiments

Model Size Ablation. Table 8 verifies that RSFT+CED is effective across model scales from 1.7B to 8B parameters.

The absolute improvement ranges from +3.1pp (1.7B) to +6.3pp (4B) to +4.9pp (8B). The 4B model occupies a sweet spot: large enough to generate diverse high-quality candidates but small enough that SFT alone leaves substantial room for improvement. The 8B model achieves the highest absolute performance (rel_F1=0.491) but has smaller relative gains because its SFT baseline is already stronger. EDCR decreases consistently across all scales (-5.2 to -7.3pp), confirming the generality of CED-based evidence improvement.

Re-DocRED. Table 9 shows results on Re-DocRED, which corrects $\sim$25% false-negative annotations.

The RSFT+CED improvement (+5.55pp on Re-DocRED vs. +6.26pp on DocRED) is preserved, demonstrating that CED is not exploiting annotation noise. All methods benefit from cleaner annotations, with EDCR decreasing on Re-DocRED because some previously labeled distractors are reclassified as gold evidence.

top-k Ablation. Table 10 examines the effect of CED filtering strictness.

Strict filtering (k=1, 2,718 samples) outperforms relaxed filtering (k=3, 7,856 samples) across all metrics. The additional samples from k=3 have systematically lower CED scores (mean 0.523 vs. 0.668), introducing distractor-heavy training examples that dilute the quality signal. This confirms the "less is more" principle for RSFT: the SFT stage already provides broad coverage, and RSFT needs only a small number of high-quality samples to calibrate evidence citation behavior.

NLI Model Ablation. Table 11 shows that replacing DeBERTa-v3-base (86M) with DeBERTa-v3-large (304M) yields marginal improvements (+0.38pp rel_F1), confirming that CED's effectiveness stems from the contrastive design rather than the NLI model's capacity.

Table 8: Model size ablation (DocRED dev set). RSFT+CED improves consistently across 1.7B–8B.

Model	Method	rel_F1	Evi-F1	EDCR↓	△rel_F1
Qwen3-1.7B	SFT	0.341	0.753	0.746	—
	RSFT+CED	0.372	0.778	0.693	+3.12pp
Qwen3-4B	SFT	0.406	0.798	0.712	—
	RSFT+CED	0.468	0.816	0.639	+6.26pp
Qwen3-8B	SFT	0.442	0.813	0.689	—
	RSFT+CED	0.491	0.837	0.619	+4.89pp

Table 9: DocRED vs. Re-DocRED (dev set). RSFT+CED gains are preserved on cleaner annotations.

Dataset	Method	rel_F1	Evi-F1	EDCR↓
DocRED	SFT	0.406	0.798	0.712
	RSFT+CED	0.468	0.816	0.639
Re-DocRED	SFT	0.453	0.811	0.689
	RSFT+CED	0.509	0.838	0.602

The top-1 candidate selection agrees between base and large models on 89% of documents; the 11% disagreements produce only marginal downstream differences, lowering the computational barrier for adopting FREIGE.

Baseline Comparison with Evidence Methods. Table 12 contextualizes FREIGE against encoder-based evidence methods.

5 Discussion

Why RSFT Succeeds Where RL Fails. The evidence–extraction decoupling phenomenon reveals that document-level IE presents unique challenges for on-policy RL: (1) QLoRA's rank-limited parameters are easily overwritten by RL gradients that preferentially optimize the more gradient-accessible evidence dimension; (2) evidence-quality rewards provide stronger gradients than extraction-accuracy signals, causing asymmetric optimization; (3) the combinatorial output space of multiple triples with evidence makes credit assignment intractable for policy-gradient methods. RSFT bypasses all three issues by using standard supervised learning on curated data, inheriting the stability of SFT while benefiting from CED's quality signal.

Limitations of EDCR. Entity-sharing is an operationalized proxy for distractor sentences, not a perfect definition. The auto–human gap of +6.8pp (Table 7) indicates that some entity-sharing sentences provide partial contextual support. Additionally, EDCR does not capture non-entity-sharing distractors (sentences that are irrelevant but happen not to mention target entities). We recommend using EDCR as a complementary diagnostic alongside Evi-F1, not as a replacement.

Extraction Gap vs. Encoders. FREIGE's primary limitation is the extraction accuracy gap compared to encoder-based methods (rel_F1 0.455 vs. DREEAM's $\sim$0.631). This gap is inherent to the generative paradigm under current model scales and is not specific to our method—LMRC (LLaMA-3.1-8B) achieves 0.607 with 2$\times$ parameters and no evidence output. As LLM capabilities scale, this gap is expected to narrow, making evidence generation increasingly valuable.

Table 10: top-k ablation for CED filtering (DocRED dev set). Stricter filtering (k=1) yields higher quality despite fewer training samples.

top-k	n_{train}	**Mean CED**	**rel_F1**	**Evi-F1**	**EDCR↓**
1	2,718	0.668	**0.468**	**0.816**	**0.639**
3	7,856	0.523	0.452	0.803	0.671

Table 11: NLI model ablation for CED scoring. Larger NLI model provides marginal gains.

NLI Model	**Params**	**Disc. Acc**	**RSFT rel_F1**	**RSFT Evi-F1**
DeBERTa-v3-base	86M	0.815	0.468	0.816
DeBERTa-v3-large	304M	0.847	0.472	0.819
Δ	—	+0.032	+0.38pp	+0.33pp

6 Conclusion

We presented FREIGE, the first LLM-based document-level RE framework that jointly generates relation triples and verifiable sentence-level evidence. Our core technical contribution, CED, provides a contrastive evidence scoring mechanism that is effective as an offline data quality filter but causes catastrophic forgetting when used as an online RL reward—revealing a previously unreported evidence–extraction decoupling phenomenon in GRPO+QLoRA for structured IE. The proposed EDCR metric, validated by human annotation (r=0.62), captures a systematic failure mode of evidence citation invisible to standard Evi-F1. Experiments across three model scales, two dataset variants, multiple RL paradigms, and extensive ablations demonstrate the robustness of our approach.

Table 12: Evidence baseline comparison (dev set). FREIGE is the first generative LLM method with explicit evidence; the Evi-F1$_J$ gap vs. encoders is driven by extraction accuracy, not evidence quality.

Method	**Type**	**rel_F1**	**Evi-F1$_J$**	**Evidence Form**
DREEAM	Encoder	0.631	0.572	Attention threshold
SAIS	Encoder	0.610	0.554	Attention threshold
LMRC	Gen. LLM	0.607	—	None
SFT baseline	Gen. LLM	0.406	0.322	Explicit citation
FREIGE	Gen. LLM	0.468	0.381	Explicit citation